

a double spend without a central authority illustrates the primary advantage of using a blockchain to maintain the underlying ledger.

The initial cryptocurrency model is the Bitcoin blockchain, which functions almost exclusively as a payment network, with the capabilities of storing and transacting bitcoins across the globe in real time with no intermediaries or censorship. This is powerful value proposition gives bitcoin its value. Even though its network effects are strong, some competitors in the cryptocurrency space offer enhanced functionality.

THE SMART CONTRACT PLATFORM

A crucial ingredient of DeFi is a *smart contract* platform, which goes beyond a simple payments network such as Bitcoin and enhances the chain's capabilities. Ethereum is the primary example. A smart contract is code that can create and transform arbitrary data or tokens on top of the blockchain to which it belongs. Powerfully, it allows the user to trustlessly encode rules for any type of transaction and even create scarce assets with specialized functionality. Many of the clauses of traditional business agreements could be shifted to a smart contract, which not only would enumerate but also algorithmically enforce those clauses. Smart contracts go beyond finance to include gaming, data stewardship, and supply chain.

Ethereum charges a *gas fee* for every transaction – similar to how driving a car takes a certain amount of gas, which